

Exercises

Vector Space and Subspace

1. Define vector space with examples and non-examples.
2. Define subspace with examples and non-examples.
3. Write the first and second fundamental theorem of subspace.
4. Show that $W = \{(a, b, c) \in \mathbb{R}^3 : a + b + c = 0\}$ is a subspace of $\mathbb{R}^3$.
5. Show that $W = \{(a, b, c) \in \mathbb{R}^3 : a + b = 0\}$ is a subspace of $\mathbb{R}^3$.
6. Show that $T = \{(a, b, c, d) \in \mathbb{R}^4 : 2a - 3b + 5c - d = 0\}$ is a subspace of $\mathbb{R}^4$.
7. Determine whether the following subset of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$. Justify your answer using the subspace test.
 - i) $S = \{(x, y, 1) \in \mathbb{R}^3 : x, y \in \mathbb{R}\}$
 - ii) $T = \{(x, y, z) \in \mathbb{R}^3 : 3x + 2y - z = 0, x, y, z \in \mathbb{R}\}$
 - iii) $S = \{(x, y, z) \in \mathbb{R}^3 : x \geq 0, x, y, z \in \mathbb{R}\}$
 - iv) $T = \{(x, y, z) \in \mathbb{R}^3 : x, y, z \in \mathbb{R}, x^2 + y^2 + z^2 \leq 1\}$
8. Examine whether the following subset of $\mathbb{R}^4$ is a subspace of $\mathbb{R}^4$ and justify your answer.
 - i) $S = \{(a, b, c, d) \in \mathbb{R}^4 : a, b, c, d \in \mathbb{R}, a \geq b\}$
 - ii) $T = \{(a, b, c, 0) \in \mathbb{R}^4 : a, b, c \in \mathbb{R}\}$

Linear Combination of Vectors

1. Show that the vector $\underline{v}$ can be expressed as a linear combination of the vectors $\underline{v}_1, \underline{v}_2, \underline{v}_3$, where $\underline{v} = (5, 9, 5)$ and $\underline{v}_1 = (2, 1, 4), \underline{v}_2 = (1, -1, 3), \underline{v}_3 = (3, 2, 5)$.
2. Express the vector $\underline{v}$ as a linear combination of the vectors $\underline{v}_1, \underline{v}_2$, where $\underline{v} = (2, -5, 4)$ and $\underline{v}_1 = (1, -3, 2), \underline{v}_2 = (2, -1, 1)$.
- 3.

Which of the following are linear combinations of

$\mathbf{u} = (0, -2, 2)$ and $\mathbf{v} = (1, 3, -1)$?

- (a) $(2, 2, 2)$ (b) $(0, 4, 5)$ (c) $(0, 0, 0)$

4.

Express the following as linear combinations of $\mathbf{u} = (2, 1, 4)$,
 $\mathbf{v} = (1, -1, 3)$, and $\mathbf{w} = (3, 2, 5)$.

- (a) $(-9, -7, -15)$ (b) $(6, 11, 6)$ (c) $(0, 0, 0)$

5.

Which of the following are linear combinations of

$$A = \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}?$$

$$(a) \begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix} \quad (b) \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} -1 & 5 \\ 7 & 1 \end{bmatrix}$$

6.

Express the vector $\mathbf{v}$ as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$, where

$$\mathbf{v} = \begin{bmatrix} 5 \\ -7 \\ 3 \end{bmatrix}, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

That is, determine c_1, c_2 such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$$

7.

Determine whether the vector

$$\mathbf{v} = \begin{bmatrix} 4 \\ 1 \\ 7 \end{bmatrix}$$

can be expressed as a linear combination of

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

8.

Express

$$\mathbf{v} = \begin{bmatrix} 5 \\ 1 \\ 7 \\ -2 \end{bmatrix}$$

as a linear combination of

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix}.$$

Linear Independence and Dependence of Vectors

1. Show that the vectors $(1, 1, -1)$, $(1, 0, 2)$ and $(1, 1, 1)$ in $\mathbb{R}^3$ are linearly independent.
2. Show that the vectors $(1, -2, 1)$, $(2, 1, -1)$ and $(7, -4, 1)$ in $\mathbb{R}^3$ are linearly dependent.
3. Show that the vectors $(1, -1, 2)$, $(3, -5, 1)$, $(2, 7, 8)$ and $(-1, 1, 1)$ in $\mathbb{R}^3$ are linearly dependent.
- 4.

In each part, determine whether the vectors are linearly independent or are linearly dependent in R^3 .

- (a) $(-3, 0, 4)$, $(5, -1, 2)$, $(1, 1, 3)$
- (b) $(-2, 0, 1)$, $(3, 2, 5)$, $(6, -1, 1)$, $(7, 0, -2)$

5.

In each part, determine whether the vectors are linearly independent or are linearly dependent in R^4 .

- (a) $(3, 8, 7, -3)$, $(1, 5, 3, -1)$, $(2, -1, 2, 6)$, $(4, 2, 6, 4)$
- (b) $(3, 0, -3, 6)$, $(0, 2, 3, 1)$, $(0, -2, -2, 0)$, $(-2, 1, 2, 1)$

6.

In each part, determine whether the matrices are linearly independent or dependent.

(a) $\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$ in M_{22}

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ in M_{23}

7. Determine whether the functions $f(t) = \sin t$, $g(t) = \cos t$ and $h(t) = t$ are linearly independent or dependent.

Linear Spans

1. Determine whether the following vectors span $\mathbb{R}^3$ or not.

i) $(1, 2, 1), (2, 1, 0), (1, -1, 2)$

ii) $(1, -2, 1), (2, -1, 1), (2, 3, 1)$

2.

Test whether the following set of vectors is sufficient to generate $\mathbb{R}^3$:

$$(1, 0, 2), \quad (2, 1, 3), \quad (0, 1, -1).$$

3.

Determine whether the vectors

$$(1, 2, 3), \quad (2, -1, 1)$$

span $\mathbb{R}^3$.

4.

Can every vector in $\mathbb{R}^3$ be expressed as a linear combination of

$$\mathbf{v}_1 = (1, 0, 2), \quad \mathbf{v}_2 = (2, 1, 1)?$$

Give a reason for your answer.

5.

Determine whether the following four vectors span $\mathbb{R}^3$:

$$(1, 0, 1), \quad (0, 1, 1), \quad (1, 1, 0), \quad (2, 1, 1).$$

6.

Do the vectors

$$(1, 2, 0), \quad (0, 1, 1), \quad (2, 3, 1), \quad (1, 0, -1)$$

span $\mathbb{R}^3$?

7.

Consider the set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

Determine whether

$$\text{span}(S) = \mathbb{R}^3.$$

8.

Determine whether the following vectors span $\mathbb{R}^4$:

$$(1, 0, 2, 1), \quad (0, 1, 1, 2), \quad (2, 1, 3, 4).$$

9.

Let

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \\ 3 \end{bmatrix} \right\}.$$

Determine whether S spans $\mathbb{R}^4$.

Basis and Dimension of vector space

- Determine whether the following set of vectors are a basis of $\mathbb{R}^3$ or not.
 - $\{(1,2,0), (0,5,7), (-1,1,3)\}$
 - $\{(1,3-4), (1,4,-3), (2,3,-11)\}$
 - $\{(1,1,1), (1,0,1)\}$
- Determine whether $\{(1,1,1,1), (1,2,3,2), (2,5,6,4), (2,6,8,5)\}$ form a basis of $\mathbb{R}^4$.

Linear Transformation

- Test whether the transformation defined as follows are linear or not.
 - $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3: T(x, y, z) = (x + 2y, x - 3z, 2x + y + z)$
 - $T: \mathbb{R}^3 \rightarrow \mathbb{R}^5: T(x, y, z) = (x - y, 0, x - z, y, 0)$
 - $T: \mathbb{R}^2 \rightarrow \mathbb{R}: T(x, y) = xy$
 - $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2: T(x, y) = (x - y, x - z)$
 - $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2: T(x, y, z) = (x + 1, y + z)$
 - $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3: T(x, y, z) = (x + y, x + 1, z + x)$

Eigenvalues and Eigenvectors

- Find the eigenvalues and associated eigenvectors of the following matrices.

i) $A = \begin{pmatrix} 1 & 4 \\ 9 & 1 \end{pmatrix}$

ii) $A = \begin{pmatrix} -14 & 12 \\ -20 & 17 \end{pmatrix}$

iii) $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$

$$\text{iv)} \quad A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

2. Find a matrix P that diagonalizes the matrix $A = \begin{pmatrix} 1 & 0 \\ 6 & -1 \end{pmatrix}$.

3. Find a non-singular matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix, for

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}.$$

4. Diagonalize the following matrix by finding a suitable matrix P :

$$A = \begin{pmatrix} -3 & 1 & -1 \\ -7 & 5 & -1 \\ -6 & 6 & -2 \end{pmatrix}.$$

5. Find the eigenvalues and associated eigenvectors of the matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$. Is the matrix A diagonalizable? Justify your answer.

Cayley- Hamilton Theorem

1. Verify Cayley-Hamilton theorem for the following matrices. Hence, find their inverse.

$$\text{i)} \quad A = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$$

$$\text{ii)} \quad A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$$

$$\text{iii)} \quad A = \begin{pmatrix} 1 & -2 & 0 \\ 0 & -8 & 1 \\ 3 & 1 & -1 \end{pmatrix}$$